

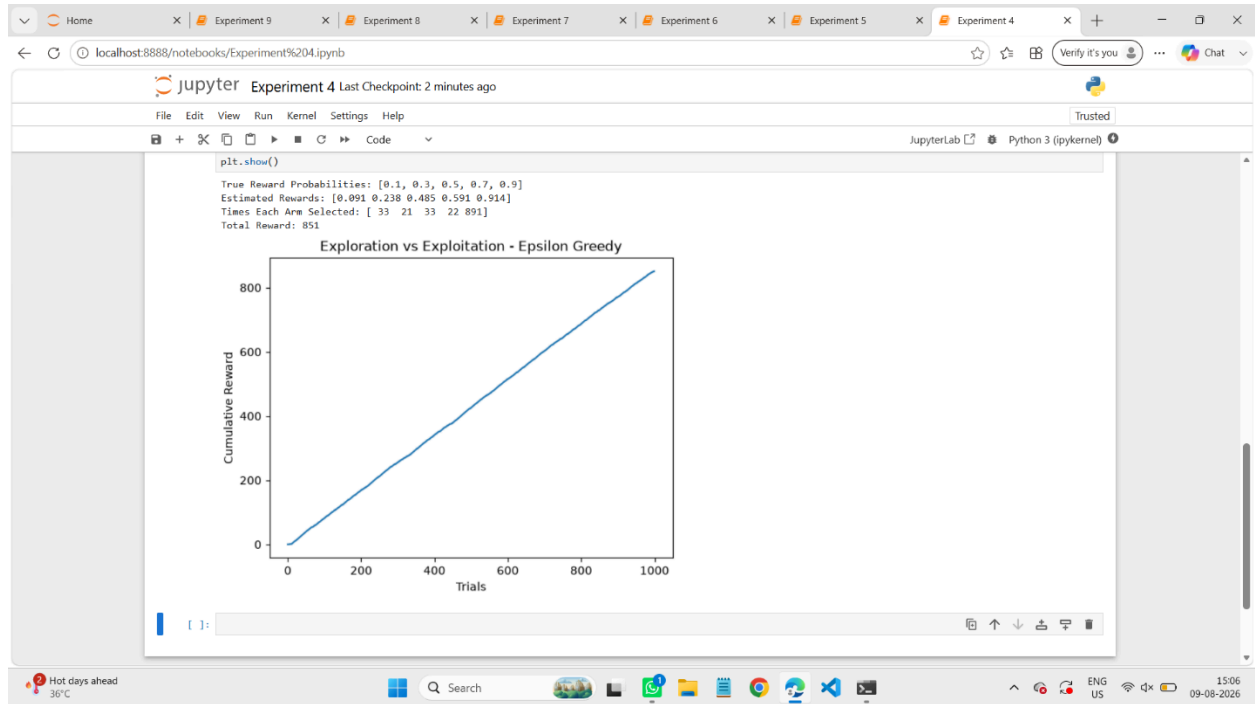
Experiment 4 RL

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Course Code: MLA0305

Slot: B



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JupyterLab Python 3 (ipykernel)

```
if np.random.random() < epsilon:
    # Explore: choose a random arm
    arm = np.random.randint(n_arms)
else:
    # Exploit: choose the arm with highest estimated reward
    arm = np.argmax(Q)

# Generate reward
reward = 1 if np.random.random() < true_rewards[arm] else 0

# Update count
N[arm] += 1

# Update estimated reward
Q[arm] += (reward - Q[arm]) / N[arm]

rewards.append(reward)

# Display results
print("True Reward Probabilities:", true_rewards)
print("Estimated Rewards:", np.round(Q, 3))
print("Times Each Arm Selected:", N.astype(int))
print("Total Reward:", sum(rewards))

# Plot cumulative reward
plt.plot(np.cumsum(rewards))
plt.xlabel("Trials")
plt.ylabel("Cumulative Reward")
plt.title("Exploration vs Exploitation - Epsilon Greedy")
plt.show()

True Reward Probabilities: [0.1, 0.3, 0.5, 0.7, 0.9]
Estimated Rewards: [0.091 0.238 0.485 0.591 0.914]
Times Each Arm Selected: [ 33  21  33  22 891]
Total Reward: 851
```

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JupyterLab Python 3 (ipykernel)

```
[1]: import numpy as np
import matplotlib.pyplot as plt

# Number of arms
n_arms = 5

# Number of trials
n_trials = 1000

# Exploration probability
epsilon = 0.1

# True reward probabilities of each arm
true_rewards = [0.1, 0.3, 0.5, 0.7, 0.9]

# Estimated rewards
Q = np.zeros(n_arms)

# Number of times each arm is selected
N = np.zeros(n_arms)

rewards = []

for i in range(n_trials):

    # Exploration or exploitation
    if np.random.random() < epsilon:
        # Explore: choose a random arm
        arm = np.random.randint(n_arms)
    else:
        # Exploit: choose the arm with highest estimated reward
        arm = np.argmax(Q)

    # Generate reward
```

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